ACE

Engineering College

An Autonomous Institution

Ankushapur (V), Ghatkesar (M), Medchal Dist. – 501 301

Assignment 1(A)

Last Date for Submission: 13-03-2026 **Course**: CS603PC – Artificial Intelligence

Total Marks: 2.5

Question 1: Define the following terms in your own words with suitable examples: Agent, Agent Function, Agent Program, Rational Agent, Autonomy, and Environment.

Question 2: Write pseudocode for the following agents discussed in the Artificial Intelligence- Modern Approach Textbook by Russel Norvig:

a) Simple Reflex Agent.

b) Goal-Based Agent.

Explain briefly how goal-based agents differ from reflex agents.

Question 3: Consider the 8-puzzle problem. Explain the concept of heuristic search. Describe the Manhattan Distance heuristic and explain why it is admissible.

Question 4: Compare Breadth-First Search (BFS) and Depth-First Search (DFS) with respect to:

- Completeness.
- Time Complexity.
- Space Complexity.

Provide one example scenario where BFS performs better than DFS.

Note:

- All assignments must be submitted on or before the mentioned date; late submissions will not be accepted, and students who fail to submit will be awarded 0 marks.

Presentation Guidelines:

- Use A4 sheets. Write your Name, Roll Number, Section, and Department clearly on the first page.
- Maintain logical flow and neat presentation throughout.

ACE

Engineering College

An Autonomous Institution

Ankushapur (V), Ghatkesar (M), Medchal Dist. – 501 301

Assignment 1(B)

Last Date for Submission: 13-03-2026 **Course:** CS603PC – Artificial Intelligence

Total Marks: 2.5

Question 1: Discuss how the standard approach to game playing in Artificial Intelligence (minimax search with evaluation functions) can be applied to physical games such as tennis, pool, or croquet.

Question 2: Explain the Map Coloring problem as a Constraint Satisfaction Problem (CSP). Identify the variables, domains, and constraints using the Australia map example.

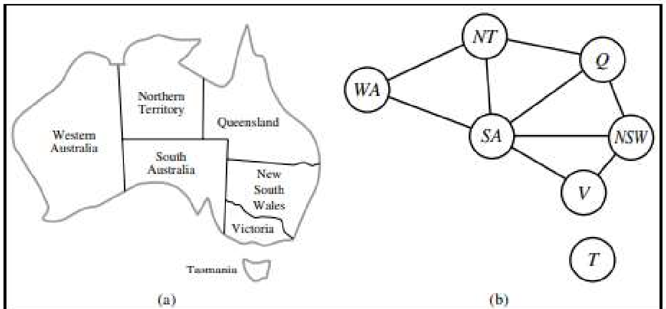

Figure 1 (a) The principal states and territories of Australia. Coloring this map can be viewed as a constraint satisfaction problem (CSP). The goal is to assign colors to each region so that no neighboring regions have the same color. (b) The map-coloring problem represented as a constraint graph.

Question 3: Solve the cryptarithmetic puzzle SEND + MORE = MONEY using constraint satisfaction. Explain the constraints used in the solution process.

Question 4: Explain the following CSP techniques with a small example:

a) Forward Checking

b) Minimum Remaining Values (MRV) heuristic

Note:

- All assignments must be submitted on or before the mentioned date; late submissions will not be accepted, and students who fail to submit will be awarded 0 marks.

Presentation Guidelines:

- Use A4 sheets. Write your Name, Roll Number, Section, and Department clearly on the first page.
- Maintain logical flow and neat presentation throughout.

ACE

Engineering College

An Autonomous Institution

Ankushapur (V), Ghatkesar (M), Medchal Dist. – 501 301

Assignment 2

Last Date for Submission: 30-03-2026 **Course**: CS603PC – Artificial Intelligence

Total Marks: 5

Q1. Knowledge Representation

Concepts: Define Ontological Engineering and explain the representation of Categories, Objects, and Events (including fluents).

Mental States: Distinguish between Mental Events and Mental Objects in an agent's knowledge base.

Process: List the key steps in the Knowledge Engineering process as defined in AIMA.

Q2. Planning with GraphPlan

Construction: Draw a Planning Graph for a 2-block "Blocks-World" problem (A on B $\rightarrow B$ on A) up to Level S_2.

Mutex: Identify and define the three types of Mutual Exclusions (Inconsistent Effects, Interference, Competing Needs).

Extraction: Explain how GraphPlan performs Backward Search to extract a valid plan.

Q3. Probabilistic Inference (Bayes' Rule)

The Formula: State Bayes' Rule for $P(H \mid E)$.

Application: A disease has a 1% prior prevalence. A test has a 95% sensitivity and a 10% false positive rate. Calculate the posterior probability $P(Disease \mid Positive)$.

Briefly explain why the result is lower than the sensitivity.

Q4. Bayesian Networks & Dempster-Shafer

Semantics: Explain how a Bayesian Network factorizes a full joint distribution using conditional independence.

Factorization: Show the joint probability equation for a 4-node network

(e.g., $Cloudy \rightarrow Rain \rightarrow WetGrass$).

Dempster-Shafer: Define the Belief (Bel) and Plausibility (Pl) functions and explain how they represent "ignorance" differently than standard Bayes.

Note:

- All assignments must be submitted on or before the mentioned date; late submissions will not be accepted, and students who fail to submit will be awarded 0 marks.

Presentation Guidelines:

- Use A4 sheets. Write your Name, Roll Number, Section, and Department clearly on the first page.
- Maintain logical flow and neat presentation throughout.